

Infron vs OpenRouter Routing, Cache, Cost, and Streaming Performance A/B Test Report

Abstract and Executive Outline

Keywords: Prompt Caching; A/B Testing; Provider Routing; Cache Affinity; Latency; Throughput; Cost Attribution; qwen3.6-flash

Abstract

This report evaluates `qwen/qwen3.6-flash` on Infron and OpenRouter across cache reuse, observed cost, throughput, E2E latency, and Streaming TTFT under Prompt Caching workloads.

The main findings are: Infron and OpenRouter tie on Cache hit rate in all routing modes; Infron leads Observed cost in all routing modes; OpenRouter leads Throughput in all routing modes; Infron leads E2E latency in 1/4 routing modes; OpenRouter leads in 3/4; 0/4 tie; OpenRouter leads Streaming TTFT in all routing modes.

Overall, Infron's cross-mode strengths are observed cost, while OpenRouter's cross-mode strengths are throughput, Streaming TTFT, E2E latency. Platform choice should be driven by workload objective rather than a single headline metric.

Figure 0: Normalized Capability Radar

The five radar axes represent throughput, price, E2E latency, Streaming TTFT, and cache hit rate. Every metric is normalized to a 0–100 score, with farther outward meaning better.

Thick solid lines show platform–level contours; translucent lines and points show individual routing modes.

Conclusion Overview: Core Metric Winners by Routing Mode

Based on strict A/B paired samples. Blue represents Infron, orange represents OpenRouter; gold cells mark the winner for the routing objective.

Throughput OpenRouter wins 4/4 Max advantage 36.27%, higher is better	Observed cost Infron wins 4/4 Max advantage 144.66%, lower is better	E2E latency OpenRouter wins 3/4 Max advantage 18.05%, lower is better	Streaming TTFT OpenRouter wins 4/4 Max advantage 10.94%, lower is better	Cache hit rate Tie in 4/4 modes Max advantage 0.00%, higher is better
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Routing mode	Throughput objective	Cost objective	Latency objective	TTFT objective	Cache result
Throughput First throughput	OpenRouter advantage 14.85%	Infron advantage 131.07%	OpenRouter advantage 1.06%	OpenRouter advantage 9.37%	Tie tie
Price First price	OpenRouter advantage 36.27%	Infron advantage 144.66%	Infron advantage 18.05%	OpenRouter advantage 7.93%	Tie tie
Latency First latency	OpenRouter advantage 13.58%	Infron advantage 130.10%	OpenRouter advantage 3.46%	OpenRouter advantage 10.94%	Tie tie
TTFT First ttft	OpenRouter advantage 12.27%	Infron advantage 130.59%	OpenRouter advantage 0.50%	OpenRouter advantage 7.71%	Tie tie

Executive Outline

Dimension	Conclusion	Evidence
Controls	First/second <code>usage.prompt_tokens</code> deltas are limited to 50 tokens within each <code>sort/group/round</code> pair.	Methods and data quality
Cache reuse	Infron and OpenRouter tie on Cache hit rate in all routing modes	Overall metrics and mechanism section
Observed cost	Infron leads Observed cost in all routing modes	Overall metrics and provider drill-down
Performance	OpenRouter leads Throughput in all routing modes; Infron leads E2E latency in 1/4 routing modes; OpenRouter leads in 3/4; 0/4 tie; OpenRouter leads Streaming TTFT in all routing modes	Charts and statistical tests
Attribution boundary	Claims use observable response telemetry: usage, cost, TTFT, latency, provider fields, and cache tokens.	Provider/Route drill-down
Business meaning	Long-context, RAG-prefix, agent-tool, and high-frequency template workloads should evaluate cache rate, cost, first-token latency, and E2E latency together.	Discussion and conclusion

Routing–Mode Conclusions

Routing mode	Objective	Cache winner	Cost winner	Throughput winner	E2E latency winner	TTFT winner	Interpretation
Throughput First	Maximize output capacity per unit time	Tie	Infron	OpenRouter	OpenRouter	OpenRouter	OpenRouter leads overall (3/5 metrics)
Price First	Minimize request and token cost	Tie	Infron	OpenRouter	Infron	OpenRouter	Mixed result
Latency First	Minimize full–response waiting time	Tie	Infron	OpenRouter	OpenRouter	OpenRouter	OpenRouter leads overall (3/5 metrics)
TTFT First	Minimize streaming first–token time	Tie	Infron	OpenRouter	OpenRouter	OpenRouter	OpenRouter leads overall (3/5 metrics)

1. Introduction: Background, Questions, and Contributions

LLM inference–platform behavior is shaped by provider routing, prompt caching, streaming response handling, cost attribution, and fallback policy. This report treats the platform as an observable system and measures speed, cost, cache reuse, and first–token experience through strict A/B pairs.

1.1 Research Hypotheses

Hypothesis	Statement	Validation metric
H1	Stronger provider/cache affinity improves token–level cache hit rate for repeated stable prefixes.	Second–request cache–read tokens and token cache hit rate
H2	Higher cache hit rate can reduce observed cost, but does not necessarily reduce TTFT or E2E latency.	Observed cost, average TTFT, average request latency
H3	Different routing sorts change provider selection and produce different cost/throughput/latency frontiers.	Provider distribution, throughput, latency, cost

1.2 Contributions

- Uses response–returned `usage.prompt_tokens` as the input–token control while allowing small cross–platform accounting variance up to 50 tokens.
- Extends prompt–caching evaluation to cost, throughput, E2E latency, TTFT, provider distribution, reasoning telemetry, and paired statistical tests.
- Scopes every claim to observable response telemetry instead of private routing internals.

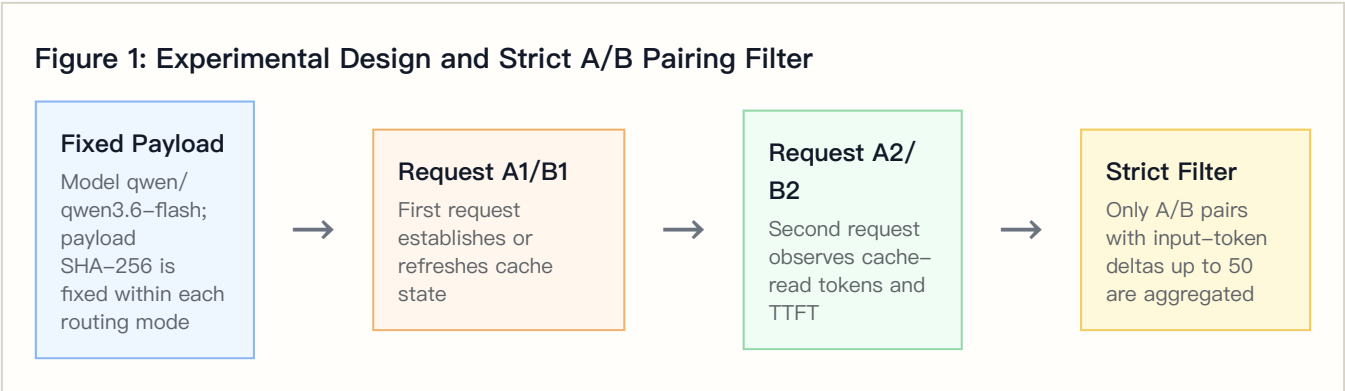
2. Experimental Design, Dataset, and Controls

2.1 Dataset Construction

The dataset is `controlled_cache_probe`, covering 4 routing sorts, 2 platforms, 4 groups, and 50 rounds per group. Each round sends identical first/second Chat Completions requests; the second request observes cache-read tokens, TTFT, and E2E latency.

The built-in business templates represent stable long-context workloads such as RAG support, agent tool instructions, marketing automation, and code review.

2.2 Controlled Variables



Controlled-variable rule: within the same `sort/group/round`, both platforms must have first/second `usage.prompt_tokens` deltas no greater than 50 tokens. Total Input Tokens are computed from response-returned usage, not local tokenizer estimates.

2.3 Metric Definitions

Metric	Definition	Direction
Call cache hit rate	Share of second requests with <code>cache_read_tokens > 0</code>	Higher is better
Token cache hit rate	Second-request cache-read tokens / second-request prompt tokens	Higher is better
Observed cost	Sum of first + second request usage/cost values	Lower is better
Throughput	Completion tokens / request latency seconds	Higher is better
E2E latency	Full response latency per request	Lower is better
TTFT	Streaming first-token arrival time	Lower is better
Reasoning telemetry	Reasoning tokens from response usage	Used to explain latency, throughput, and cost

3. Experimental Environment and Data Quality

Item	Configuration
Model	qwen/qwen3.6-flash
Provider model IDs	infron: qwen/qwen3.6-flash ; openrouter: qwen/qwen3.6-flash
Platforms	Infron and OpenRouter
API protocol	/v1/chat/completions
Routing modes	Throughput First, Price First, Latency First, TTFT First
Groups	4
Rounds per group	50
Workers	24
Request mode	Streaming Chat Completions with TTFT collection
Reasoning / thinking control	No explicit reasoning/thinking parameter; model and platform defaults are preserved
Prompt length tiers	short ≈1500, medium ≈8000, long ≈32000
Excluded records	0

4. Results: Overall Metrics and Main Findings

Routing mode	Platform	Strict pairs	Total Input Tokens	Token cache hit rate	Observed cost	Throughput	E2E latency	Streaming TTFT
Throughput First	Infron	200	7076798	0.00%	\$0.66515400	73.81 tok/s	5564.89 ms	2990.30 ms
Throughput First	OpenRouter	200	7076798	0.00%	\$1.53695738	84.77 tok/s	5506.76 ms	2734.06 ms
Price First	Infron	200	7076798	0.00%	\$0.66773100	73.79 tok/s	5742.80 ms	3163.75 ms
Price First	OpenRouter	200	7076798	0.00%	\$1.63363762	100.55 tok/s	6779.32 ms	2931.36 ms
Latency First	Infron	200	7076798	0.00%	\$0.66874800	72.18 tok/s	5941.66 ms	3185.38 ms
Latency First	OpenRouter	200	7076798	0.00%	\$1.53876638	81.98 tok/s	5742.78 ms	2871.39 ms
TTFT First	Infron	200	7076798	0.00%	\$0.66694300	72.03 tok/s	5827.88 ms	3133.68 ms
TTFT First	OpenRouter	200	7076798	0.00%	\$1.53790350	80.86 tok/s	5798.80 ms	2909.38 ms

4.1 Tail Latency and Statistical Tests

Tail percentiles expose risk that averages hide.

Routing mode	Platform	P50 latency	P95 latency	P99 latency	P50 TTFT	P95 TTFT	P99 TTFT
Throughput First	Infron	5384.23 ms	7888.00 ms	10098.55 ms	2703.02 ms	5774.26 ms	7524.32 ms
Throughput First	OpenRouter	5389.78 ms	8197.47 ms	9694.70 ms	2450.31 ms	5306.60 ms	6762.72 ms
Price First	Infron	5579.58 ms	7960.64 ms	9239.78 ms	2803.26 ms	5878.20 ms	6628.75 ms
Price First	OpenRouter	5486.27 ms	8289.54 ms	13258.32 ms	2643.99 ms	5180.83 ms	7259.27 ms
Latency First	Infron	5778.95 ms	8858.83 ms	10550.89 ms	2861.22 ms	6124.34 ms	7664.66 ms
Latency First	OpenRouter	5587.00 ms	8346.71 ms	10663.67 ms	2648.19 ms	5270.19 ms	7915.19 ms
TTFT First	Infron	5651.84 ms	8051.44 ms	10052.71 ms	2849.49 ms	5754.67 ms	6776.20 ms
TTFT First	OpenRouter	5609.28 ms	8472.12 ms	10521.16 ms	2674.53 ms	5408.93 ms	7228.92 ms

Mean deltas use bootstrap 95% CIs; p-values use paired sign-flip permutation tests.

Routing mode	Metric	Mean delta	95% CI	p-value	Pairs	Interpretation
Throughput First	Latency: OpenRouter – Infron	–116.26 ms	–511.61 ms to 275.59 ms	0.5519	200	Positive means lower Infron latency
Throughput First	TTFT: OpenRouter – Infron	–512.49 ms	–775.22 ms to –270.95 ms	<0.001	200	Positive means lower Infron TTFT
Throughput First	Throughput: Infron – OpenRouter	–10.0164 tok/s	–12.7168 tok/s to –7.3521 tok/s	<0.001	200	Positive means higher Infron throughput
Throughput First	Cost: OpenRouter – Infron	\$0.00435902	\$0.00386004 to \$0.00484105	<0.001	200	Positive means lower Infron cost
Throughput First	Token Cache Hit: Infron – OpenRouter	0.00 pp	0.00 pp to 0.00 pp	1.0000	200	Positive means higher Infron cache hit
Price First	Latency: OpenRouter – Infron	2073.05 ms	–308.79 ms to 6552.23 ms	0.4516	200	Positive means lower Infron latency
Price First	TTFT: OpenRouter – Infron	–464.78 ms	–795.48 ms to –68.77 ms	0.0037	200	Positive means lower Infron TTFT
Price First	Throughput: Infron – OpenRouter	–10.1569 tok/s	–13.6306 tok/s to –6.7554 tok/s	<0.001	200	Positive means higher Infron throughput
Price First	Cost: OpenRouter – Infron	\$0.00482953	\$0.00401691 to \$0.00597316	<0.001	200	Positive means lower Infron cost
Price First	Token Cache Hit: Infron – OpenRouter	0.00 pp	0.00 pp to 0.00 pp	1.0000	200	Positive means higher Infron cache hit
Latency First	Latency: OpenRouter – Infron	–397.78 ms	–808.49 ms to 34.07 ms	0.0605	200	Positive means lower Infron latency
Latency First	TTFT: OpenRouter – Infron	–627.98 ms	–869.90 ms to –361.98 ms	<0.001	200	Positive means lower Infron TTFT

Routing mode	Metric	Mean delta	95% CI	p-value	Pairs	Interpretation
Latency First	Throughput: Infron – OpenRouter	-9.6672 tok/s	-12.3581 tok/s to -6.9634 tok/s	<0.001	200	Positive means higher Infron throughput
Latency First	Cost: OpenRouter – Infron	\$0.00435009	\$0.00385312 to \$0.00483373	<0.001	200	Positive means lower Infron cost
Latency First	Token Cache Hit: Infron – OpenRouter	0.00 pp	0.00 pp to 0.00 pp	1.0000	200	Positive means higher Infron cache hit
TTFT First	Latency: OpenRouter – Infron	-58.17 ms	-500.22 ms to 389.32 ms	0.8113	200	Positive means lower Infron latency
TTFT First	TTFT: OpenRouter – Infron	-448.60 ms	-672.50 ms to -226.91 ms	<0.001	200	Positive means lower Infron TTFT
TTFT First	Throughput: Infron – OpenRouter	-8.8558 tok/s	-11.4885 tok/s to -6.3066 tok/s	<0.001	200	Positive means higher Infron throughput
TTFT First	Cost: OpenRouter – Infron	\$0.00435480	\$0.00384752 to \$0.00484407	<0.001	200	Positive means lower Infron cost
TTFT First	Token Cache Hit: Infron – OpenRouter	0.00 pp	0.00 pp to 0.00 pp	1.0000	200	Positive means higher Infron cache hit

4.2 Reasoning / Thinking Control Check

This run does not explicitly set reasoning/thinking parameters and keeps model/platform defaults; this table records reasoning telemetry under default behavior.

Routing mode	Platform	Reasoning tokens	Avg reasoning tokens/request	Reasoning requests	Avg first reasoning token	Avg TTFT	Avg E2E latency
Throughput First	Infron	161099	402.7475	400	2990.30 ms	2990.30 ms	5564.89 ms
Throughput First	OpenRouter	183518	458.7950	400	2734.06 ms	2734.06 ms	5506.76 ms
Price First	Infron	166302	415.7550	400	3183.28 ms	3163.75 ms	5742.80 ms
Price First	OpenRouter	269458	673.6450	400	2931.36 ms	2931.36 ms	6779.32 ms
Latency First	Infron	168357	420.8925	400	3203.15 ms	3185.38 ms	5941.66 ms
Latency First	OpenRouter	185126	462.8150	400	2871.39 ms	2871.39 ms	5742.78 ms
TTFT First	Infron	164706	411.7650	400	3151.36 ms	3133.68 ms	5827.88 ms
TTFT First	OpenRouter	184359	460.8975	400	2909.38 ms	2909.38 ms	5798.80 ms

4.3 API Protocol Compatibility Matrix

This run uses `/v1/chat/completions` ; this table records success response, usage, cost, and cache telemetry coverage for both platforms under that protocol.

API protocol	Endpoint	Platform	Pairs	Requests	Success rate	Usage coverage	Token usage coverage	Cost coverage	Cache telemetry coverage
chat_completions	/v1/chat/completions	Infron	800	1600	100.00%	100.00%	100.00%	100.00%	100.00%
chat_completions	/v1/chat/completions	OpenRouter	800	1600	100.00%	100.00%	100.00%	100.00%	100.00%

5. Result Visualizations by Routing Mode

The following charts are driven by the strict-paired summary data and present routing-mode, platform, cost, cache, E2E latency, Streaming TTFT, and upstream provider differences.

5.1 Core Metric Chart Overview

The charts below are generated from the same strict-paired summary data.

**Figure 3–E2:
Normalized
capability
contour**

Five metrics are normalized to 0–100, where higher is better.

**Figure 3–E3:
Cost–cache
efficiency plane**

X-axis is token cache hit rate; Y-axis is observed total cost.

**Figure
3–
E1:
Routing–
mode
core
metric
comparison**

Infron and OpenRouter are shown side by side by routing mode.

Figure 3–E4: E2E latency and Streaming TTFT

Solid lines show E2E latency; dashed lines show Streaming TTFT.

Figure 3–E5: Upstream provider distribution

Provider shares explain cache–domain and performance differences.

Routing–Mode Drill–Down Charts

The horizontal axis shows relative advantage: blue to the right indicates Infron advantage, orange to the left indicates OpenRouter advantage.

Figure 4–E1: Throughput First core metric comparison

Direction and magnitude of relative advantage across five metrics.

Figure 4–E2: Price First core metric comparison

Direction and magnitude of relative advantage across five metrics.

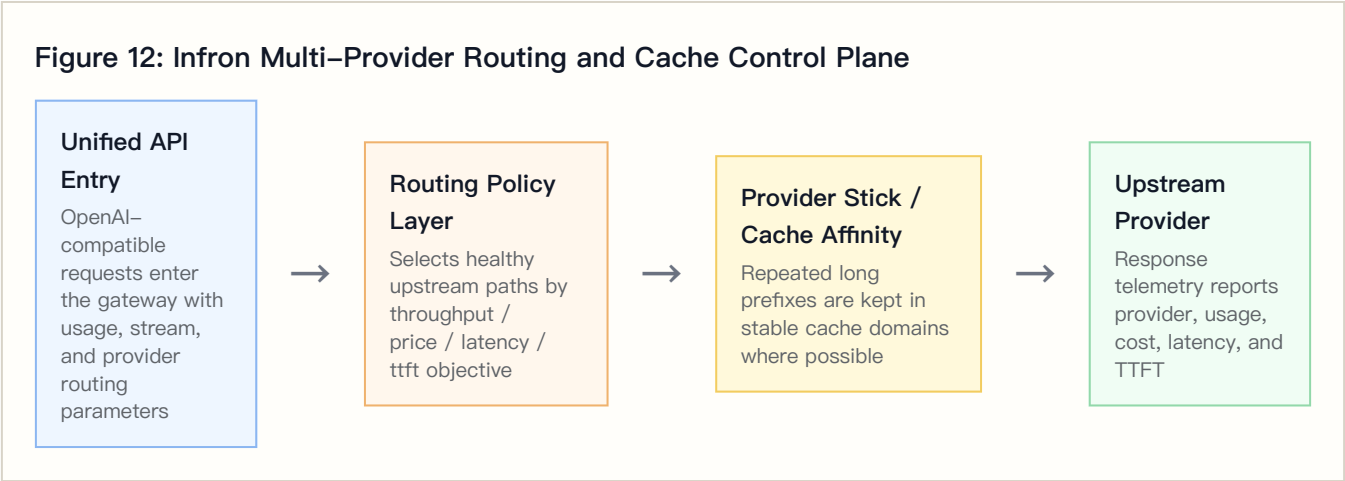
Figure 4–E3: Latency First core metric comparison

Direction and magnitude of relative advantage across five metrics.

Figure 4–E4: TTFT First core metric comparison

Direction and magnitude of relative advantage across five metrics.

6. Infron Technical Architecture and Cache/Cost Mechanism



6.1 Multi-Provider Routing and Observable Control Plane

Requests enter a unified API, and the routing layer selects healthy upstream paths according to throughput, price, latency, or ttft objectives. Whether stable long prefixes stay in the same cache domain directly affects second-request cache-read tokens.

6.2 Provider Stick and Cache Affinity

Provider stick is cache affinity, not a permanent provider lock. Its goal is to reduce cache-domain fragmentation while respecting health and SLA constraints.

6.3 Cost-Control Path

Mechanism	Cache-rate effect	Cost effect	Observable signal
Stable prefix detection	Repeated prefixes are more likely to hit cache	Reduces repeated prefill cost	Payload SHA-256 and second-request cache-read tokens
Provider stick / cache affinity	Reduces cross-domain cache fragmentation	Avoids repeated cache warm-up	Provider distribution and token cache hit rate
Health checks and fallback	Protects availability while sometimes sacrificing cache	Reduces failure cost	HTTP status, provider distribution, tail latency
Cost-aware routing	Prefers lower-cost paths under constraints	Reduces total and per-round cost	Observed cost, cost breakdown coverage, cache-read tokens

7. Provider/Route Drill-Down

Routing mode	Platform	Total requests	Attributed requests	Provider distribution
Throughput First	Infron	400	400	alibaba/us 400
Throughput First	OpenRouter	400	400	Alibaba 400
Price First	Infron	400	400	alibaba/eu 400

Routing mode	Platform	Total requests	Attributed requests	Provider distribution
Price First	OpenRouter	400	400	Alibaba 400
Latency First	Infron	400	400	alibaba/eu 400
Latency First	OpenRouter	400	400	Alibaba 400
TTFT First	Infron	400	400	alibaba/eu 400
TTFT First	OpenRouter	400	400	Alibaba 400

Upstream Provider Detail Distribution

Routing mode	Platform	Upstream provider	Requests	Share	first/second	Covered rounds	Avg TTFT	Avg latency	Prompt tokens	Completion tokens	Re to
Throughput First	Infron	alibaba/us	400	100.00%	200/200	200	2990.30 ms	5564.89 ms	7076798	164299	16
Throughput First	OpenRouter	Alibaba	400	100.00%	200/200	200	2734.06 ms	5506.76 ms	7076798	186718	18
Price First	Infron	alibaba/eu	400	100.00%	200/200	200	3163.75 ms	5742.80 ms	7076798	169498	16
Price First	OpenRouter	Alibaba	400	100.00%	200/200	200	2931.36 ms	6779.32 ms	7076798	272656	26
Latency First	Infron	alibaba/eu	400	100.00%	200/200	200	3185.38 ms	5941.66 ms	7076798	171555	16
Latency First	OpenRouter	Alibaba	400	100.00%	200/200	200	2871.39 ms	5742.78 ms	7076798	188326	18
TTFT First	Infron	alibaba/eu	400	100.00%	200/200	200	3133.68 ms	5827.88 ms	7076798	167904	16
TTFT First	OpenRouter	Alibaba	400	100.00%	200/200	200	2909.38 ms	5798.80 ms	7076798	187559	18

7.1 Cache–Rate and Cost Divergence Drill–Down

This table combines cache, cost, provider distribution, and reasoning telemetry to explain routing–mode differences.

Routing mode	Cache–hit delta	Infron cost multiple	Infron top path	OpenRouter top path	Reasoning token delta	Main attribution
Throughput First	+0.00 pp	0.43x	alibaba/us 100.00%	Alibaba 100.00%	–22419	Cache is tied; Infron leads cost
Price First	+0.00 pp	0.41x	alibaba/eu 100.00%	Alibaba 100.00%	–103156	Cache is tied; Infron leads cost
Latency First	+0.00 pp	0.43x	alibaba/eu 100.00%	Alibaba 100.00%	–16769	Cache is tied; Infron leads cost

Routing mode	Cache-hit delta	Infron cost multiple	Infron top path	OpenRouter top path	Reasoning token delta	Main attribution
TTFT First	+0.00 pp	0.43x	alibaba/eu 100.00%	Alibaba 100.00%	-19653	Cache is tied; Infron leads cost

8. Stratified Results: Prompt-Length Cache Performance

This section aggregates second-request cache-read tokens, token-level cache hit rate, observed cost, E2E latency, and Streaming TTFT by prompt-length tier. Bold cells mark the advantaged side within each tier.

Prompt-Length Tier Overview

Prompt length tier	Target tokens	Platform	Pairs	Second prompt tokens	Second cache read tokens	Token cache hit rate	Observed cost	Avg E2E latency	Avg TTFT
short	1500	Infron	268	533808	0	0.00%	\$0.20024900	5141.94 ms	1822.18 ms
short	1500	OpenRouter	268	533808	0	0.00%	\$0.49131562	5543.65 ms	1875.74 ms
medium	8000	Infron	268	2767588	0	0.00%	\$0.57990300	5598.44 ms	2859.92 ms
medium	8000	OpenRouter	268	2767588	0	0.00%	\$1.41749925	6160.97 ms	2667.63 ms
long	32000	Infron	264	10852200	0	0.00%	\$1.88842400	6579.64 ms	4696.29 ms
long	32000	OpenRouter	264	10852200	0	0.00%	\$4.33845000	6169.29 ms	4059.15 ms

Prompt Length x Routing Mode Cache Hit Rate

Prompt length tier	Routing mode	Infron	OpenRouter	Winner
short	Throughput First	0.00%	0.00%	tie
short	Price First	0.00%	0.00%	tie
short	Latency First	0.00%	0.00%	tie
short	TTFT First	0.00%	0.00%	tie
medium	Throughput First	0.00%	0.00%	tie
medium	Price First	0.00%	0.00%	tie
medium	Latency First	0.00%	0.00%	tie
medium	TTFT First	0.00%	0.00%	tie

Prompt length tier	Routing mode	Infron	OpenRouter	Winner
long	Throughput First	0.00%	0.00%	tie
long	Price First	0.00%	0.00%	tie
long	Latency First	0.00%	0.00%	tie
long	TTFT First	0.00%	0.00%	tie

9. Stratified Results: Group–Level Stability

Throughput First

Platform	Group	Rounds	Successful rounds	Token cache hit rate	Observed cost	P95 latency	P95 TTFT
Infron	1	50	50	0.00%	\$0.16326100	7821.91 ms	6430.84 ms
Infron	2	50	50	0.00%	\$0.16977300	7954.44 ms	5653.84 ms
Infron	3	50	50	0.00%	\$0.16917000	8138.58 ms	5353.94 ms
Infron	4	50	50	0.00%	\$0.16295000	7391.40 ms	4728.24 ms
OpenRouter	1	50	50	0.00%	\$0.37724175	8292.57 ms	5352.14 ms
OpenRouter	2	50	50	0.00%	\$0.39415088	7404.65 ms	4885.65 ms
OpenRouter	3	50	50	0.00%	\$0.38795963	8188.80 ms	5414.37 ms
OpenRouter	4	50	50	0.00%	\$0.37760512	8633.33 ms	4731.43 ms

Price First

Platform	Group	Rounds	Successful rounds	Token cache hit rate	Observed cost	P95 latency	P95 TTFT
Infron	1	50	50	0.00%	\$0.16450300	8079.94 ms	5675.71 ms
Infron	2	50	50	0.00%	\$0.17084900	7683.09 ms	4854.19 ms
Infron	3	50	50	0.00%	\$0.16762600	7694.15 ms	5812.01 ms
Infron	4	50	50	0.00%	\$0.16475300	8025.84 ms	6338.45 ms
OpenRouter	1	50	50	0.00%	\$0.37843425	7282.62 ms	4654.63 ms
OpenRouter	2	50	50	0.00%	\$0.39319350	7613.72 ms	5401.41 ms
OpenRouter	3	50	50	0.00%	\$0.48523950	8832.67 ms	5036.16 ms
OpenRouter	4	50	50	0.00%	\$0.37677037	9436.64 ms	4799.51 ms

Latency First

Platform	Group	Rounds	Successful rounds	Token cache hit rate	Observed cost	P95 latency	P95 TTFT
Infron	1	50	50	0.00%	\$0.16562200	7966.88 ms	4976.04 ms

Platform	Group	Rounds	Successful rounds	Token cache hit rate	Observed cost	P95 latency	P95 TTFT
Infron	2	50	50	0.00%	\$0.17128000	8859.04 ms	6261.04 ms
Infron	3	50	50	0.00%	\$0.16828400	8171.99 ms	6026.08 ms
Infron	4	50	50	0.00%	\$0.16356200	9035.32 ms	6471.12 ms
OpenRouter	1	50	50	0.00%	\$0.37881113	8511.14 ms	5280.34 ms
OpenRouter	2	50	50	0.00%	\$0.39254775	9054.91 ms	6652.89 ms
OpenRouter	3	50	50	0.00%	\$0.38827575	7756.80 ms	4678.76 ms
OpenRouter	4	50	50	0.00%	\$0.37913175	7733.10 ms	5005.44 ms

TTFT First

Platform	Group	Rounds	Successful rounds	Token cache hit rate	Observed cost	P95 latency	P95 TTFT
Infron	1	50	50	0.00%	\$0.16466400	8687.30 ms	5778.38 ms
Infron	2	50	50	0.00%	\$0.17059200	8033.11 ms	5711.28 ms
Infron	3	50	50	0.00%	\$0.16777200	7976.26 ms	5577.07 ms
Infron	4	50	50	0.00%	\$0.16391500	7859.73 ms	5568.88 ms
OpenRouter	1	50	50	0.00%	\$0.37969425	8715.35 ms	5687.67 ms
OpenRouter	2	50	50	0.00%	\$0.39302813	8369.42 ms	5592.18 ms
OpenRouter	3	50	50	0.00%	\$0.38917462	8862.39 ms	5416.21 ms
OpenRouter	4	50	50	0.00%	\$0.37600650	7425.02 ms	4723.04 ms

10. Discussion: Business Value, Boundaries, and Engineering Implications

Business decisions should not rely on one metric. Stable long-context and high-frequency template workloads should prioritize cache rate and cost; realtime interaction must constrain TTFT and E2E latency; batch processing often prioritizes throughput and failure cost.

Routing mode	Business objective	Observed result	Scenarios	Caveat
Throughput First	Maximize output capacity per unit time	OpenRouter leads overall (3/5 metrics)	Batch generation, offline summaries, backend processing	First-token and E2E response are faster; check cache and cost
Price First	Minimize request and token cost	Mixed result	High-frequency templates, support automation, marketing, RAG prefixes	Good for throughput-first tasks; constrain cost and cache separately
Latency First	Minimize full-response waiting time	OpenRouter leads overall (3/5 metrics)	Online chat, agent chains, IDE/ writing assistants, realtime tools	First-token and E2E response are faster; check cache and cost

Routing mode	Business objective	Observed result	Scenarios	Caveat
TTFT First	Minimize streaming first-token time	OpenRouter leads overall (3/5 metrics)	Streaming chat, realtime copilots, first-screen feedback	First-token and E2E response are faster; check cache and cost

11. Conclusion

Routing-Mode Conclusions

Routing mode	Objective	Cache winner	Cost winner	Throughput winner	E2E latency winner	TTFT winner	Interpretation
Throughput First	Maximize output capacity per unit time	Tie	Infron	OpenRouter	OpenRouter	OpenRouter	OpenRouter leads overall (3/5 metrics)
Price First	Minimize request and token cost	Tie	Infron	OpenRouter	Infron	OpenRouter	Mixed result
Latency First	Minimize full-response waiting time	Tie	Infron	OpenRouter	OpenRouter	OpenRouter	OpenRouter leads overall (3/5 metrics)
TTFT First	Minimize streaming first-token time	Tie	Infron	OpenRouter	OpenRouter	OpenRouter	OpenRouter leads overall (3/5 metrics)

12. Limitations, Missing Data, and Next Experiments

Limitation	Impact	Next step	Current handling
Full routing trace	Cannot prove every provider choice, fallback, and retry path hop by hop	Add provider routing trace, decision logs, and fallback reasons	Use only returned provider fields and provider distribution
Longer time window	4x50 observes short-window stability but not day-level drift	Add soak tests and repeated windows	Scope conclusions to this run
Production corpus	Built-in templates do not cover every workload distribution	Use sanitized production-stratified corpora	Discuss representative long-context templates only
Cost-field consistency	Cost coverage and semantics may differ by platform	Reconcile with billing and provider cost breakdown	Use only explicitly returned cost fields

13. Reproducibility Appendix

Artifact	Path
Summary	<code>export/qwen3_6_flash_all_experiments/routing_sort_cache_cost_ab_4x50_stream_ttft_reasoning_default_length_tiers_chat_completions_summary.json</code>

Artifact	Path
Paired dataset	<code>export/qwen3_6_flash_all_experiments/ routing_sort_cache_cost_ab_4x50_stream_ttft_reasoning_default_length_tiers_chat_completions benchmark_pairs.csv</code>
Request-level dataset	<code>export/qwen3_6_flash_all_experiments/ routing_sort_cache_cost_ab_4x50_stream_ttft_reasoning_default_length_tiers_chat_completions benchmark_requests.jsonl</code>
Filtered structured records	<code>export/qwen3_6_flash_all_experiments/ routing_sort_cache_cost_ab_4x50_stream_ttft_reasoning_default_length_tiers_chat_completions records.json</code>
Excluded-record audit	<code>export/qwen3_6_flash_all_experiments/ routing_sort_cache_cost_ab_4x50_stream_ttft_reasoning_default_length_tiers_chat_completions records_excluded.json</code>
Test source	<code>tests/test_rerun_routing_sort_cache_cost_ab.py</code>
Benchmark runner source	<code>scripts/rerun_routing_sort_cache_cost_ab.py</code>
HTML report renderer source	<code>scripts/render_glm52_deepseek_style_report.py</code>
Dataset reference	<code>business_representative</code> built-in representative business templates; request-level export is <code>benchmark_requests</code>